GRADE 12 / Mathematical Literacy / English

Interpret financial growth

01 Understand

Learning goals: Distinguish simple and compound interest • Calculate annual compound growth • Explain purchasing power

Simple interest is calculated on the original principal. Compound interest adds interest to the balance before the next period.

For annual compounding use $A = P(1 + i)^n$, with i as a decimal and n in years. A different compounding period needs a matching rate and number of periods.

Inflation describes rising prices. A higher money balance does not guarantee greater purchasing power; compare growth over the same period.

R1 000 at an illustrative 10% per year

Year	Simple balance	Compound balance
0	R1 000	R1 000
1	R1 100	R1 100
2	R1 200	R1 210
3	R1 300	R1 331

02 Follow the reasoning

Calculate the balance on R2 000 at 8% compounded annually for two years.

1. P = 2 000, i = 0.08 and n = 2.

2. $A = 2\,000(1.08)^2 = R2\,332.80$.

3. Interest earned = 2 332.80 − 2 000 = R332.80. This example excludes fees and tax.

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Practise with purpose

03 Try before checking

Use paper or your notebook. Show your reasoning. For written responses, use evidence and explain your choices. Suggested time: 15–20 minutes.

1. Calculate simple interest on R2 000 at 8% for two years.

2. A basket rises from R500 to R540. Calculate inflation for that basket.

3. If income stays R5 000 while prices rise, what happens to purchasing power?

04 Check and correct

1. I = 2 000 × 0.08 × 2 = R320; total R2 320.

2. 40 ÷ 500 × 100 = 8%.

3. It falls: the same income buys fewer goods, assuming the price changes apply to those goods.

Common mistake to avoid

8% must be entered as 0.08 in a growth formula, not as 8.

Reflect and revisit

Identify one correction in your own words. Attempt a similar question tomorrow without looking at the guide. If you are still unsure, take your attempt to a teacher or tutor.

Coverage and sources

CAPS topic focus: Finance: interest and inflation

DBE CAPS, Grades 10–12: Mathematical Literacy. Section 3 content and skills. This is a focused supplement, not a complete syllabus or prescribed literature replacement.

Open the official DBE curriculum reference

Content and visuals: original SmartBridge FET. AI-assisted authoring; subject-teacher and language-specialist review pending. Report an issue with the pack ID matlit-12.